Sean Chien, Ph.D.

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EDUCATION

Ph.D.	Texas A&M University – College Station, Texas Biology (*Advisor*: Heath Blackmon)	2022 – 2026
M.S.	Estern Illinois University – Charleston, Illinois Biological Sciences (*Advisor*: Zhiwei Liu)	2020 – 2022

PUBLICATIONS

Peer-Reviewed Publications

1. Hu, Z., **Chien, S.**, Laird, K., & Blackmon, H. (2026). Gene Density Homogeneity is Disrupted by Repetitive DNA Expansion. *PLOS Biology*, 21(10), e0361209. https://doi.org/10.1371/journal.pone.0361209

2. **Chien, S.**, Le, M.-H., Blackmon, H., & Huang, J.-P. (2026). Differential Demographic Trajectories and Polygenic Adaptation to Monsoon Seasonality in Sky Island Hercules Beetle Populations of the American Southwest. *Molecular Ecology*, 35(18), e70551. https://doi.org/10.1111/mec.70551

3. **Chien, S.**, & Blackmon, H. (2026). Chromosomal Rearrangements: Tempo and Mode of Karyotype Evolution in Scarabaeoidea. *Journal of Evolutionary Biology*, Volume 39, Issue 6, June 2026, Pages 699-708, https://doi.org/10.1093/jeb/voag025

4. **Chien, S.**, Huang, J.-P., & Blackmon, H. (2026). Chromosome-Level Reference Genome of an Endemic, Endangered Long-Armed Scarab (*Cheirotonus formosanus*): Discovery of a Putative Y-Linked Scaffold and Demographic History. *Ecology and Evolution, 16*(4), e73483. https://doi.org/10.1002/ece3.73483

5. Heath Blackmon, Michelle Jonika, **Sean Chien**, Max Chin, Nathan, Anya, Chromartie, & Meghann McConnell. (2026). coleoguy/evobir: v3.0 (Version v3). Zenodo. https://doi.org/10.5281/zenodo.19137903

6. Sylvester, T., Hoover, Z., Hjelmen, C. E., Jonika, M. M., Blackmon, L. T., Alfieri, J. M., Johnston, J. S., **Chien, S.**, Esfandani, T., & Blackmon, H. (2024). A reference quality genome assembly for the jewel scarab *Chrysina gloriosa*, *G3 Genes |Genomes |Genetics*, 14(6), jkae084. https://doi.org/10.1093/g3journal/jkae084

Preprints

Copeland, M., McConnell, M., Barboza, A., Abraham, H. M., Alfieri, J., Arackal, S., Bernard, C. E., Bryant, K., Cast, S., **Chien, S.**, Clark, E., Cruz, C. E., Diaz, A. Y., Deiterman, O., Girish, R., Harper, K., Hjelmen, C. E., Thompson, M. J., Koehl, R., ... Blackmon, H. (2026). Dismantling chromosomal stasis across the eukaryotic tree of life. In bioRxiv (p. 2026.04.14.718287). bioRxiv. https://doi.org/10.64898/2026.04.14.718287

FELLOWSHIPS & AWARDS

Environment and Sustainability Initiative Graduate Fellow	$2,500	2026
ROOT & SHOOT Travel Award (Botany 2026 Conference)	$3,000	2026
Texas A&M University Research and Presentation Travel Award	$1,000	2025
Mark and Nancy Deyrup Visiting Scholars (Archbold Biological Station)	$4,400	2024
Society of Systematic Biologists Graduate Student Research Award	$3,000	2023

RESEARCH EXPERIENCE

Research Assistant 2022 – Present

Department of Biology, Texas A&M University
Advisor: Heath Blackmon

- Acquiring grants to secure project funding and ensure financial support for initiatives
- Designed and executed fieldwork experiments, including sample collection for *de novo* genome assemblies and population genomic projects
- Performed DNA and RNA extractions, including high molecular weight DNA extractions
- Developed and implemented custom scripts in R, Python, or Bash to extract meaningful biological insights from the database

Research Assistant 2020 – 2022

Department of Biological Sciences, Eastern Illinois University
Advisor: Zhiwei Liu
- Designed and conducted a study to understand the relationship between beetle diversity and forest management in central Illinois
- Developed skills related to experimental design, data collecting, and data analysis
- Collected and analyzed field data
- Submitted a manuscript about the study findings

TEACHING EXPERIENCE

Teaching Assistant	***Texas A&M University***	
Medical Narratives	Biology Undergraduate Program	Spring, 2025 – Present
Critical Writing in Biology	Biology Undergraduate Program	Fall, 2024
Experimental Design	Biology Graduate Program	Fall, 2023 – Spring, 2024
Introductory Biology I&II	Lower Division Instruction Program	Fall, 2022 – Spring, 2024
Teaching Assistant	***Eastern Illinois University***	
Ecology and Evolution	Department of Biological Sciences	Spring, 2022
General Biology	Department of Biological Sciences	Fall, 2021
Systematics	Department of Biological Sciences	Spring, 2021

HONORS

BioGSA Engagement Scholarship	2026
EIS Poster Presentation (co-author)– 2nd Place	2026
BioGSA Engagement Scholarship	2025
TAMU Biology Art Contest – 3rd Place	2025
SRW Oral Presentation (Graduate; Agriculture & Life Sciences) – 1st Place	2025
NATEA-Dallas Scholarship Honorable Mentions	2023
Distinguished International Student Award, Eastern Illinois University	2022
Distinguished International Student Award, Eastern Illinois University	2021
International Student Scholarship, Eastern Illinois University	2020

PRESENTATION

Student/Postdoc Research Conference (Poster)	2026
Ecological Integration Symposium (Talk)	2026
Ecological Integration Symposium (Poster)	2026
Evolution(Talk)	2025
Southeast Texas Evolutionary Genetics and Genomics (Talk)	2025
Entomological Interdisciplinary Research Symposium (Talk)	2025
Ecological Integration Symposium (Talk)	2025
Texas Genetics Society (Talk)	2025
Student Research Week (Talk)	2025
Student/Postdoc Research Conference (Poster)	2025
Ecological Integration Symposium (Poster)	2024
Student Research Week (Poster)	2024
Student/Postdoc Research Conference (Poster)	2024

Texas Genetics Society (Poster)	2024
Evolution (Poster)	2023

SERVICE & OUTREACH

Ecological Intergration Symposium - Judge	2026
Faculty search graduate student committee	2025
Open Source for Open Science workshop	2025
Ecological Intergration Symposium	2024
Darwin day	2024
R workshop (Texas Genetics Society)	2024
R workshop (Texas A&M University Biology Department)	2024

PEER REVIEW

Genome (2) *Genome Biology and Evolution* (1)

PROFESSIONAL ASSOCIATION

American Society of Naturalists	2024	Society of Systematic Biologists	2024
Society of Study of Evolution	2023	Texas Genetics Society	2023

REFERENCES

Dr. Zhiwei Liu
Professor of Biology
Department of Biological Sciences, Eastern Illinois University
Email: zliu@eiu.edu

Dr. Heath Blackmon
Associate Professor of Biology
Associate Department Head of Graduate Studies
Graduate Advisor
Department of Biology, Texas A&M University
Email: blackmon@tamu.edu

Dr. Jen-Pan Huang
Associate Research Fellow
Academia Sinica Biodiversity Research Center (Taiwan) & The Field Museum of Natural History
Email: jphuang@as.edu.tw

Dr. James Alfieri
Assistant Professor of Instruction
Department of Biology, University of Iowa
Email: james-alfieri@uiowa.edu